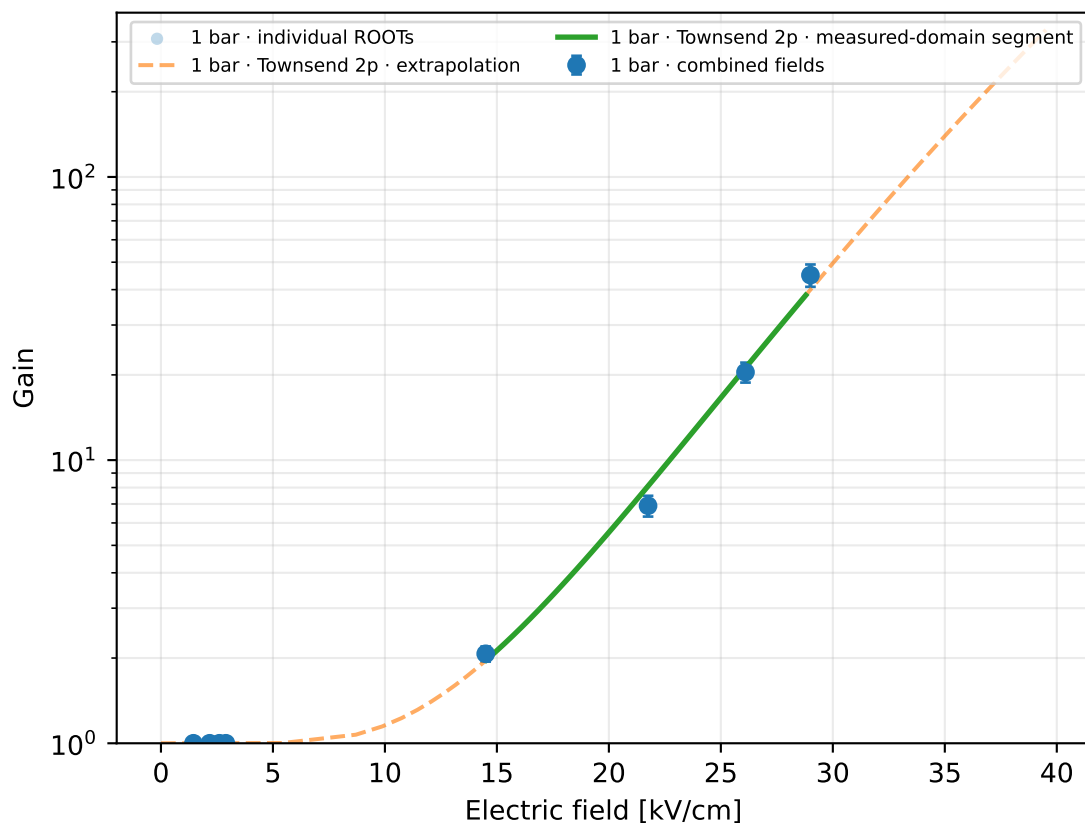
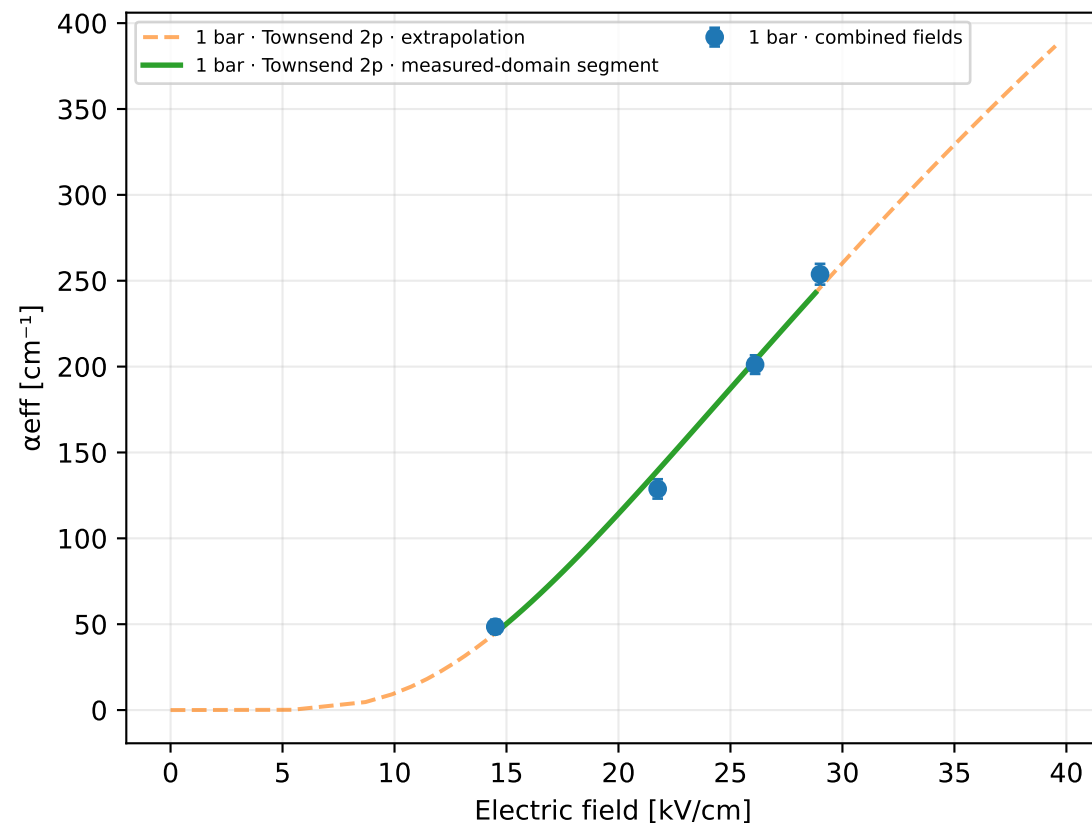


Arlso · Ar 95 % / Iso 5% · gap 0.15 mm

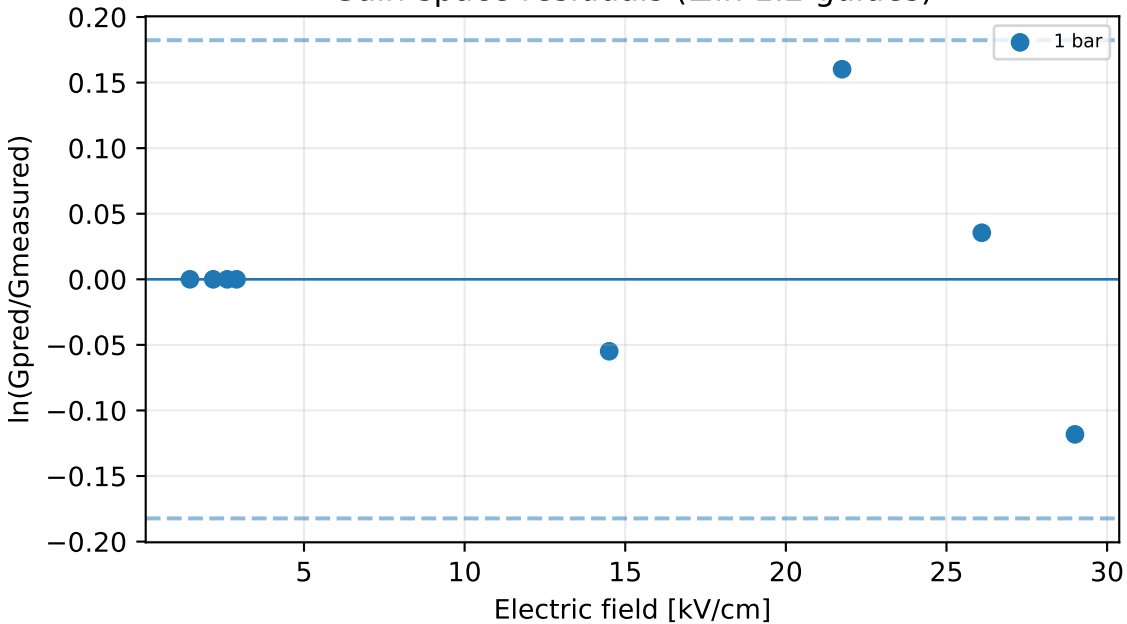
Gain vs electric field



Effective Townsend coefficient vs electric field



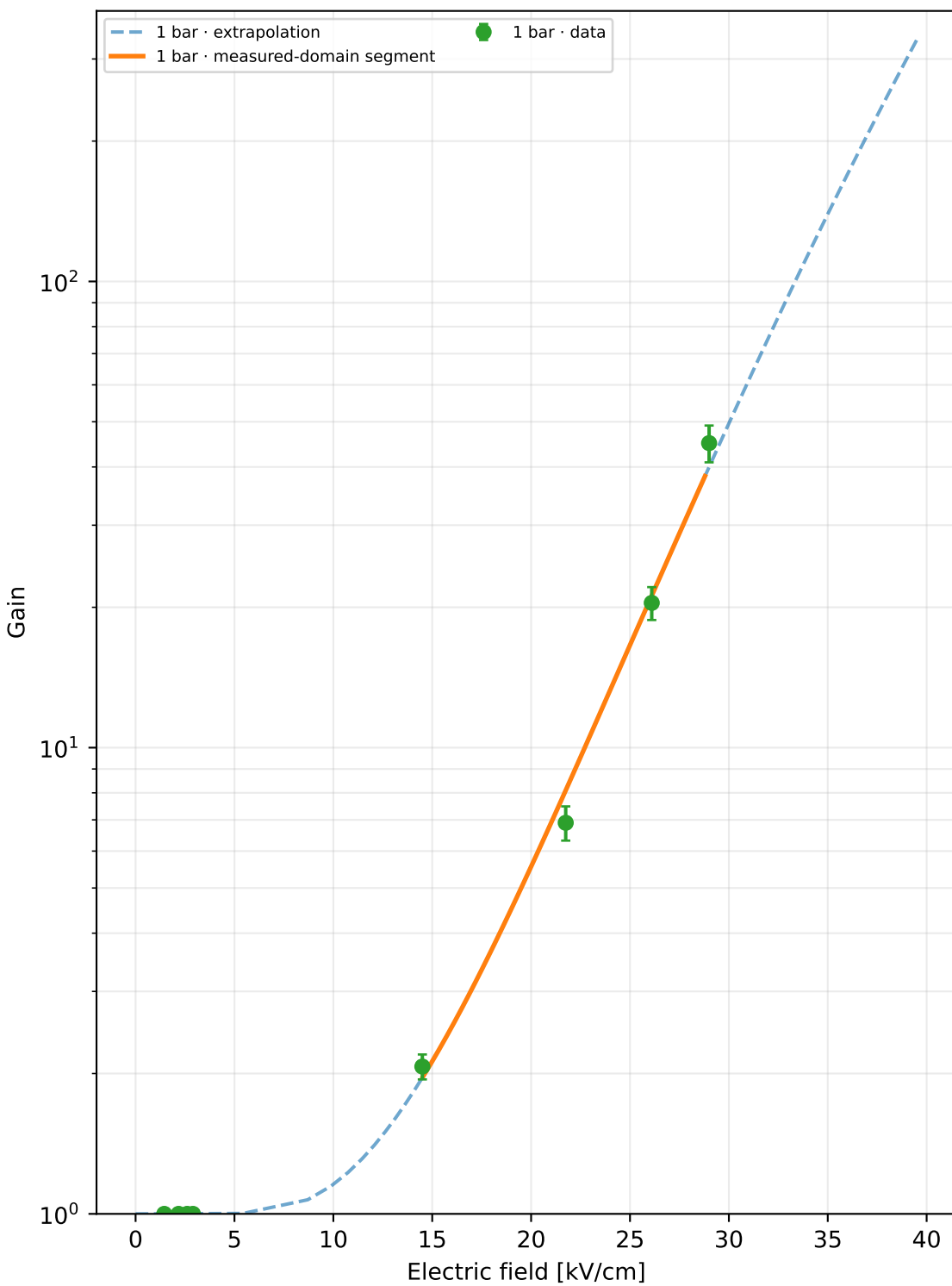
Gain-space residuals ($\pm \ln 1.2$ guides)



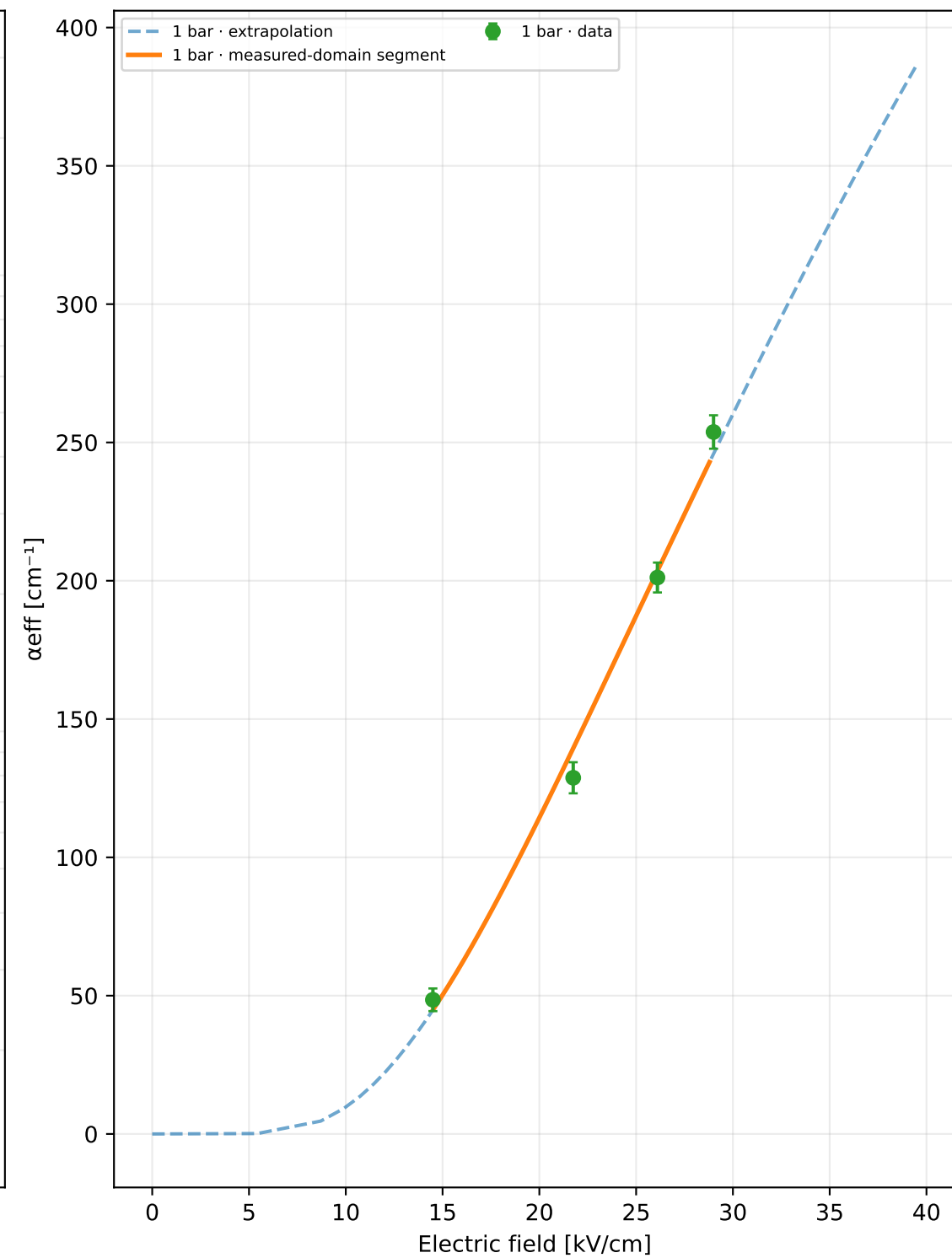
ROOT runs: 8
 Unique physical points: 8
 Pressures: 1 bar
 Selected model: Townsend 2p
 Status: rejected_poor_cross_validation
 Training RMSE: $\times 1.060$
 Cross-validation median: unavailable
 Cross-validation worst: unavailable
 Reduced-field range: 14.5–29 kV cm⁻¹ bar⁻¹
 A=1348.3, B=49.348, m=0, n=1
 Covariance condition: 213
 Warnings: poor_cross_validation, large_cross_validation_outlier
 1 bar display limit: G=331 (local inversion limit)

Extrapolation capped at $G=1e+07$ or at the local model limit

Gain vs electric field



α_{eff} vs electric field



p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm ⁻¹]	npe	runs
1	1.45	1.45	1	0	—	100	1
1	2.175	2.175	1	0	—	100	1
1	2.61	2.61	1	0	—	100	1
1	2.9	2.9	1	0	—	100	1
1	14.5	14.5	2.07	0.127	48.5032	100	1
1	21.75	21.75	6.9	0.582	128.768	100	1
1	26.1	26.1	20.44	1.66	201.166	100	1
1	29	29	45	4.08	253.777	100	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-6.73	—	—	no	poor cross validation, large cross validation outlier